

AWS Solutions Architect

Associate · SAA-C03 · Recall Poster

THE NUMBERS		pure rote — learn these first
S3 max object / single PUT		5 TB / 5 GB
■ Min duration IA / Glacier / Deep †	30 / 90 / 180 days	
S3 multipart part size		5 MB – 5 GB, max 10,000 parts
■ Glacier expedited / standard / bulk		1–5 min / 3–5 h / 5–12 h
■ Lambda max duration †	15 min (900 s)	
■ Lambda memory		128 MB – 10,240 MB
■ Lambda payload sync / async	6 MB / 256 KB	
■ Lambda default concurrency		1,000 per region
■ SQS max message		256 KB
■ SQS retention default / max	4 days / 14 days	
■ SQS visibility default / max	30 s / 12 h	
■ SQS long poll wait		20 seconds
■ Kinesis shard in / out	1 MB/s / 2 MB/s	
■ Kinesis retention default / max	24 h / 365 days	
■ DynamoDB max item		400 KB
■ DynamoDB RCU / WCU	4 KB strong read / 1 KB write	
RDS backup retention		7 days default, 0 – 35 range
■ DynamoDB LSI / GSI per table		5 / 20
■ Read replicas RDS / Aurora	5 / 15	
■ Max storage RDS / Aurora	64 TiB / 128 TiB	
VPC CIDR range		/16 down to /28
Reserved IPs / usable in a /24		5 / 251
■ Spot notice; spot & RI discount	2 min; ~90% / ~72%	
■ CloudWatch basic / detailed		5 min / 1 min

EC2		compute
Instance states		pending, running, stopping, stopped, shutting-down, terminated, rebooting
Survives reboot but not stop		instance store data + public IPv4
Stop: lost / kept		RAM & instance store / EBS volumes
Metadata endpoint, version to use		169.254.169.254, IMDSv2
Hibernate saves		RAM, to the root EBS volume
User data runs		once, at first boot, as root
System check vs instance check		AWS problem, stop/start to move host — vs your OS or config
■ Low latency / isolation / distributed	cluster / spread / partition placement group	
■ Socket or core-based licence †	Dedicated Host	
■ Steady 24/7 vs interruptible batch †=		Savings Plan or RI vs Spot
SSH user: Amazon Linux / Ubuntu		ec2-user / ubuntu (CentOS: centos, Debian: admin)
AMI, ECR image, snapshot scope		region — must be copied elsewhere
Trap: Dedicated Instance only isolates hardware per account. Only a Dedicated Host gives you the physical server and its socket and core counts.		

AUTO SCALING & LOAD BALANCING	
Default cooldown and grace period	300 seconds each
■ Recommended scaling policy	target tracking
Known spike / learned pattern	scheduled / predictive scaling
Instances killed before app boots	raise the grace period; use an ELB health check
Group flapping in and out	raise cooldown, widen the alarm threshold
Script before termination / debug one	lifecycle hook / standby
■ Define the instances	launch template (configurations are legacy)
■ ALB / NLB / GWLB layer	7 / 4 / 3
■ Static IP, UDP, millions of req/s	NLB
■ Path or host routing, dynamic container ports, Cognito auth	ALB
■ Cross-zone default †	ALB on, NLB off, Classic off
Min subnets, internet-facing	2 subnets in 2 AZs
Health check lives on / many certs	the target group / SNI

LAMBDA, CONTAINERS & APP PLATFORMS	
■ Needs longer than 15 minutes = Faster with no code change = Cold starts / starving the account 429 TooManyRequests	Fargate, ECS, Batch or Step Functions more memory (CPU scales with it) provisioned / reserved concurrency concurrency limit hit layers / dead-letter queue
■ Share dependencies / failed async	NAT gateway
■ Lambda in a VPC needs internet	your code's permissions vs ECS pulling the image
■ ECS task role vs execution role	EKS
■ Kubernetes or kubectl	Fargate / EC2 launch type
■ Containers: no servers / needs GPU	Elastic Beanstalk (Immutable or Blue/Green = zero downtime)
■ Upload code, AWS builds the stack Chef or Puppet / long batch jobs	OpsWorks / AWS Batch
■ Edge: light rewrite / calls services	CloudFront Functions / Lambda@Edge

S3		object storage
Consistency model		strong read-after-write
■ Default encryption today †	SSE-S3, applied automatically	
■ SSE-S3 / SSE-KMS / SSE-C	AWS holds all keys / you own the CMK / you send the key every request	
■ Accidental vs permanent delete		versioning / MFA Delete (root + CLI)
■ Undeletable for seven years		Object Lock, compliance mode
■ Replication requires †		versioning on both + an IAM role — and it never copies existing objects
■ Slow uploads from far away	Transfer Acceleration	
■ Browser uploads one file securely	presigned URL	
■ CloudFront reads a private bucket		Origin Access Control (OAC)
Stop all public access		S3 Block Public Access
■ Hot / unknown pattern / infrequent =	Standard / Intelligent-Tiering / Standard-IA	
■ Events can go to	SQS, SNS, Lambda, EventBridge	
■ Infrequent and re-creatable †	One Zone-IA (single AZ)	
■ Archive: ms / minutes-hours / cheapest	Glacier Instant / Flexible / Deep Archive	
Trap: One Zone-IA lives in one AZ. If the question says "must survive the loss of an Availability Zone," it is wrong, however cheap it looks.		

EBS, EFS, FSX & TRANSFER	
EBS scope +	one AZ — never spans AZs
Move a volume to another AZ	snapshot, then create from it there
Encrypt an existing volume or RDS	snapshot — copy with encryption — restore
Changeable while running	type, size, IOPS (grow only, never shrink)
Snapshots stored in	S3, incremental, region-scoped
■ General default volume	gp3 (3,000 IOPS / 125 MB/s included)
■ Guaranteed IOPS / throughput / cold	io2 / st1 / sc1 (st1 & sc1 cannot boot)
Highest IOPS, data disposable	instance store
■ Shared files, Linux, many instances ≈	EFS (NFS v4.1, regional)
■ SMB, Windows, Active Directory ≈	FSx for Windows File Server
■ HPC or ML scratch, links to S3	FSx for Lustre
■ On-prem NFS/SMB backed by S3	S3 File Gateway
■ Shrink on-prem footprint vs keep it all local	Volume Gateway cached / stored mode
■ Retire a tape library	Tape Gateway
■ Petabytes, no bandwidth vs ongoing sync ≈	Snowball / DataSync

One backup policy across services		AWS Backup
DATABASES		
■ Multi-AZ purpose †=	availability — synchronous, not readable	
■ Read replica purpose †=	read scaling — async, readable, cross-region	
■ Multi-AZ failover		automatic, 60–120 s, same endpoint
RDS gives you OS access †		no — need it? run on EC2
■ Aurora copies / failover		6 across 3 AZs / under 30 s
Restoring a snapshot gives		a new instance with a new endpoint
■ Cross-region DR, sub-second lag	Aurora Global Database	
■ Intermittent load, near-zero idle cost		Aurora Serverless v2
■ DynamoDB microsecond reads =		DAX (read-through only)
LSI vs GSI		same PK, made with the table / new PK any time, eventually consistent
■ Active-active across regions	Global Tables	
■ Every item change / expire free	DynamoDB Streams + Lambda / TTL	
■ ProvisionedThroughputExceeded		hot partition — better key, or on-demand
■ Redis vs Memcached		persistency, failover, sorted sets / simple multi-threaded cache
■ Warehouse vs ad-hoc SQL on S3		Redshift / Athena
■ Short queries stuck behind long		Redshift WLM query queues
Graph / time series / ledger		Neptune / Timestream / QLDB
■ Migrate engine, low downtime		DMS + Schema Conversion Tool
Trap: you can never read from a Multi-AZ standby. If an answer offers the standby for reporting queries it is wrong — that is what a read replica is for.		

VPC & CONNECTIVITY	
What makes a subnet public	a route to an internet gateway
■ Private instances reach the internet =	NAT gateway in a public subnet (one per AZ)
■ Security group	instance level, stateful, allow-only
■ Network ACL †	subnet level, stateless, allow + deny, in rule order
■ Block a single malicious IP †	NACL deny rule (groups cannot deny)
■ Default vs custom NACL †	allows everything / denies everything
Ephemeral port range †	1024 – 65535
■ Two VPCs vs hundreds †	peering (not transitive) / Transit Gateway
■ Expose your service to another VPC	PrivateLink + NLB
■ Gateway endpoints work with †	S3 and DynamoDB only (free)
■ Interface endpoint	everything else — an ENI with a security group
■ Hybrid in days vs steady bandwidth =	Site-to-Site VPN / Direct Connect (weeks)
■ Why traffic is being dropped	VPC Flow Logs — look for REJECT
■ Shell access, no open ports, audited =	SSM Session Manager
Trap: NACLs are stateless. An inbound rule alone is never enough — you must also allow the outbound ephemeral ports so the reply can get out.	

ROUTE 53 & THE EDGE	
A/B test, gradual migration	Weighted
Active-passive disaster recovery	Failover
Compliance, language, licensing =	Geolocation (the user's location)
Shift load between regions with a bias	Geoproximity (your resources)
Fastest region for each user =	Latency
Up to 8 healthy records	Multivalue answer
Works at the zone apex, free †	Alias (a CNAME cannot)
■ CloudFront default / max TTL	24 hours / 1 year
Health check failure threshold	3 consecutive checks
■ CloudFront certificate must live in †	us-east-1
Restrict content to paying users	signed URLs or signed cookies
■ Static staticc IPs, TCP/UDP, gaming	Global Accelerator (does not cache)

SECURITY & IDENTITY		30% of the exam
Policy evaluation order =		explicit Deny → explicit Allow → implicit deny
Never grant, only limit		SCPs, permission boundaries, session policies
■ SCP effect on the management account †		none — it is never restricted
■ App on EC2 needs S3 =		IAM role via an instance profile
■ Another account needs access		cross-account role + sts:AssumeRole
■ Workforce sign-in / mobile users		IAM Identity Center / Cognito identity pools
■ Resource-based policy must have		a Principal
■ KMS direct encryption limit		4 KB → above that, envelope encryption
■ Single-tenant, FIPS 140-2 Level 3		CloudHSM
■ Rotate DB creds / free config store		Secrets Manager / SSM Parameter Store
■ SQL injection, XSS, rate limit by IP		AWS WAF
■ DDoS Response Team, cost protection		Shield Advanced
■ GuardDuty / Inspector / Macie		detects threats / scans for CVEs / classifies PIi
Detective / Security Hub		investigates / collects all findings
■ Trusted Advisor categories		Cost, Performance, Fault tolerance, Security, Service limits
Four tag categories		Technical, Automation, Business, Security

INTEGRATION & ANALYTICS	
■ Decouple a spiky front end ⇨	SQS + auto-scale workers on queue depth
Standard queue guarantees	at-least-once, best-effort order
Exactly-once and strict order	FIFO queue (name ends .fifo)
■ Message processed twice	visibility timeout shorter than processing
■ Poison messages / empty receives	dead-letter queue / long polling
■ One message to many queues	SNS fan-out (+ filter policies)
■ Filter on event body, SaaS, replay	EventBridge
■ Coordinate steps, retries, approvals	Step Functions (SWF is legacy)
App already speaks AMQP, MQTT, JMS	Amazon MQ
■ Many consumers read the same records	Kinesis Data Streams
■ Load a stream to S3 or Redshift, no code †	Firehose (~60 s, not real time)
■ Ad-hoc SQL on S3 / cut that bill	Athena / Parquet, compress, partition
■ Hadoop & Spark / search / BI	EMR / OpenSearch / QuickSight

MONITORING, GOVERNANCE & COST	
■ CloudWatch / CloudTrail / Config	performance / who did what / configuration and compliance
Metrics needing the agent †	memory and disk space used
Count ERROR lines and alarm	metric filter on a log group
■ CloudTrail free by default	90 days of management events
Who deleted an S3 object	enable data events
Prove logs untampered / all regions	log file integrity validation / apply the trail everywhere
■ Fix a non-compliant resource	Config remediation via SSM Automation
Only required CFN section / preview	Resources / change set
■ Keep data on delete / spot drift	DeletionPolicy Retain / drift detection
Many accounts and regions / guard rails	StackSets / Control Tower + SCPs
■ Right-size / alert on spend	Compute Optimizer / AWS Budgets
■ Cut the NAT bill for S3 traffic	S3 gateway endpoint
Survive an AZ vs a region failure	multiple AZs / multiple regions

SCENARIO TRIGGERS		highest marks per minute
"minimal operational overhead"		the serverless or managed option
"highly available"		multiple AZs

How to use this Left of each row is the cue , right is the answer . Lay a ruler down the middle of a column, work down saying answers out loud, then slide it away to check. Blocks are grouped by area, so you can drill one area at a time. Amber strips are traps — where a plausible-looking answer is wrong. Marks: asked a lot asked often asked thinly — counted across 522 real practice questions, not guessed. a trap sits here. commonly reworded — see Also asked as			
30% Secure	26% Resilient	24% High-performing	20% Cost-optimized

"disaster recovery", "region outage"	multiple regions
■ "cannot be interrupted"	Spot is wrong, however cheap
■ "decouple" / "notify many subscribers"	SQS / SNS
■ "microseconds" / "sub-millisecond"	DAX / ElastiCache
■ "static IP for a firewall allow-list"	NLB or Global Accelerator
■ "Windows file share"	FSx for Windows, never EFS
"no bandwidth, petabytes, deadline"	Snow family
■ "urgent" vs "consistent bandwidth"	Site-to-Site VPN / Direct Connect
■ "do not open port 22"	Session Manager
■ "never hardcoded credentials"	an IAM role
■ "must be the only key holders"	CloudHSM, or SSE-C for S3
■ "longer than 15 minutes"	not Lambda — Fargate, Batch, Step Functions
■ "we already use Kubernetes"	EKS
■ "must not leave the AWS network"	a VPC endpoint
■ "pay per query, nothing running"	Athena
■ "petabyte warehouse, high concurrency"	Redshift
"MongoDB compatible" / "Cassandra"	DocumentDB / Keyspaces
"no central trusted authority"	Managed Blockchain
"single pane of glass" for findings	Security Hub
■ "socket or core-based licence" †	Dedicated Host
TI-E-BREAKERS, IN ORDER	
1. Managed beats self-managed. 2. Multi-AZ beats single-AZ. 3. Temporary credentials beat long-term keys. 4. A guard rail beats trusting people.	
5. Cheaper wins only if every stated requirement still holds.	
6. Reject the retired: EC2-Classic, Spot block, Classic Load Balancer, SWF.	
7. Reject the impossible: a subnet spanning AZs, an EBS volume across AZs, a CNAME at the zone apex, reading a Multi-AZ standby.	

RESILIENCE & DISASTER RECOVERY	
RTO / RPO	how long you may be down / how much data you may lose
DR strategies, cheapest first	backup and restore, pilot light, warm standby, multi-site active-active
Backup and restore	RTO in hours — only backups in the other region
Pilot light	tens of minutes — data replicating, servers off
Warm standby	minutes — a scaled-down but working copy
Multi-site active-active	near zero — full capacity, both regions serving
■ Stateless tier survives instance loss	Auto Scaling across AZs behind an ELB
■ Database survives an AZ loss	Multi-AZ
■ Database survives a region loss	cross-region replica or Aurora Global Database
■ Objects survive a region loss	S3 Cross-Region Replication
■ Static page during an outage	Route 53 failover to an S3 website
■ Guard against accidental deletion	versioning + MFA Delete + Object Lock

COST OPTIMIZATION		20% of the exam
■ Steady 24/7 workload		Savings Plans or Reserved Instances
■ Fault-tolerant, interruptible work		Spot Instances
Idle at night and weekends		scheduled scaling or an instance scheduler
Over-sized instances =		Compute Optimizer
■ Big NAT bill for S3 traffic		an S3 gateway endpoint
■ Everything sitting in S3 Standard =		lifecycle rules + Intelligent-Tiering
■ Old snapshots, unattached volumes		find with Trusted Advisor, automate with DLM
■ Incomplete multipart uploads		lifecycle rule to abort them
■ High Athena bill		Parquet, compression, partitioning
■ Spiky DynamoDB traffic		on-demand capacity mode
Cross-AZ data transfer charges		keep chatty tiers in one AZ
■ Warn before overspending	AWS Budgets + Cost Anomaly Detection	
■ See cost by team	cost allocation tags + Cost Explorer	

WHAT IS GLOBAL, REGIONAL, OR ZONAL	
■ Global services	IAM, Route 53, CloudFront, Organizations, Billing
■ Regional	EC2, S3 buckets, RDS, VPC, DynamoDB tables
■ AZ-scoped	subnets, EC2 instances, EBS volumes, ENIs, NAT gateways
■ Must be copied cross-region	AMIs, EBS & RDS snapshots, ECR images, KMS keys
A region has	3 or more AZs, independent power, cooling and network
Local Zone / Wavelength / Outposts	near a big city / in a 5G network / in your datacentre
■ Edge locations	CloudFront points of presence — far more numerous than regions

LOAD BALANCERS IN DEPTH		the single most-asked topic
■ Routes on URL path, host, header, query string	ALB (layer 7)	
■ Routes on TCP/UDP port only	NLB (layer 4)	
■ Preserves the client source IP natively	NLB (ALB uses X-Forwarded-For)	
■ Millions of requests/sec, ultra-low latency	NLB	
■ One static IP per AZ, supports Elastic IPs	NLB	
■ WebSockets, HTTP/2, gRPC	ALB	
■ Redirect HTTP to HTTPS with no app change	ALB listener rule	
■ Return a fixed response, e.g. a maintenance page	ALB listener rule	
■ Authenticate with Cognito or OIDC at the edge	ALB	
■ Route to Lambda as a target	ALB	
■ Route to IP targets, including on-premises	ALB and NLB (IP target type)	
Front a third-party firewall appliance	Gateway Load Balancer (layer 3, GENEVE)	
Health check lives on	the target group, not the listener	
TLS terminates on	the listener, with an ACM certificate	
Several certificates on one listener	SNI	
Sticky sessions	a cookie — duration or application based	
Deregistration delay default	300 s (connection draining)	
■ Spread across AZs but traffic is uneven	turn on cross-zone load balancing (off by default on NLB)	
Trap: an ALB has no static IP — always use its DNS name. If a question needs an address for a customer allow-list, the answer is NLB or Global Accelerator.		